

Math Circle!

Saturday, May 14, 2022 3:41 PM

... this puzzle illustrates a basic construction from cryptography!: The **Diffie-Hellman Protocol!** (1976).

The "true" puzzle: How do you share private information in public?

E.g.,



Which channel?



So many details!

The best we can do: ① Use mathematics cleverly to isolate key problems.

Key Trick:

Use hardness of Discrete Logarithm!

What is that? ...

② Assume your enemy is always listening, and perhaps even know your methods! (Kerchoff's Principle!)

③ Force your enemy to waste resources!

What is that? ...

WASTE RESOURCE!

Consider powers mod p , with p a prime...

Fact #1: Computing $g^x \bmod p$

(for $g, x \in \{2, \dots, p-1\}$ and p prime) is "easy," but

finding x from (g^x, g, p) is "hard".

Review: $131 \bmod 101$ means $\begin{cases} \text{subtract } 101 \\ \rightarrow \text{get } 30 \\ \text{Take remainder} \\ \text{of } 131 \div 101. \\ \text{(integer division with remainder)} \end{cases}$

We let $\mathbb{F}_{101}$ denote the finite field with 101 elements. This is just $\{1, \dots, 100\}$ endowed with $+$, $-$, $\times$, $\div$ where you reduce mod 101.

This works for any prime p , in fact, and you get

$\mathbb{F}_p$ similarly... The powers of 2, mod 101, are:

$$2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32, 2^6 = 64,$$

$$2^7 = 128 = 128 - 101 = 27 \bmod 101,$$

$$2^8 = 256 = 256 - 2 \times 101 = 54 \bmod 101, \dots$$

Funny: The powers of 2 cycle by 100s, and no smaller number!

$$\{2^0, 2^1, \dots, 2^{100} \bmod 101\} = \{1, 2, 4, 8, 16, \dots, 76, 51, 1\}.$$

$2^{99} \bmod 101$



↑

$$\{2^0, 2^1, \dots, 2^{100} \pmod{101}\} = \{1, 2, 4, 8, 16, \dots, 76, 51, 1\}.$$

$2^{100} \pmod{101}$

Puzzle: What are the powers of
95 mod 101?

$$95^0, 95^1, 95^2, \dots$$

$95^2 = (?)$... Squaring 95 is
mildly annoying ...
but $95 = 6 \pmod{101}$

$$\Rightarrow 95^2 = (-6)^2 = 36 \pmod{101}$$

$$95^3 = (-6)^3 = (-6)^2(-6) = -14 \pmod{101}$$

$$95^4 = (95^2)^2 = 36^2 = 1296 = 84 \pmod{101} = -17 \pmod{101}$$

$$95^5 = 95^4 \cdot 95 = (-17)(-6) = 1 \pmod{101}$$

→ Powers of 95 mod 101 are: 1, 95, 36, -14, -17, 1
 $\begin{array}{ccccc} & -6 & 36 & -17 & \\ & \pmod{101} & \pmod{101} & \pmod{101} & \end{array}$

Powers of 2 mod 101: There are exactly 100 distinct.

Powers of 95 mod 101: There are exactly 5 distinct.

Powers of any number ($\neq 0$) mod 101:

1 ← powers of 1
2,
5, ← powers of 95
10,
25,
50.
distinct!

10,
25,
50,
100 ← powers of 2

Puzzle/

Claim: 1 + 5 "easy" to compute powers of 2 mod 101:

e.g., you can compute 2^{35} mod 101

using just 7 multiplications mod 101 (!)

What? $2^{35} = \underbrace{2 \cdot 2 \cdot \dots \cdot 2}_{35 \text{ of 'em}}$
 3 + multi

Trick: Write 35 in binary, i.e., write

$35 = \text{sum of descending powers of 2}$

$$= 32 + 2 + 1 = 2^5 + 2^1 + 2^0$$

$$= (100011)_2.$$

Hint: $2^{35} = 2^{32+2+1} = 2^{2^5+2^1+2^0} = 2^{(2^5)} \cdot 2^{2^1} \cdot 2^{2^0}$

$$= \left(\left(\left((2^2)^2 \right)^2 \right)^2 \right)^2 \cdot 2^2 \cdot 2$$

In other words: $2^2 = 4$ ✓

$$(2^2)^2 = 16$$
 ✓

$$(2^4)^2 = ((2^2)^2)^2 = 2^{(2^3)} = 2^8 = 54 \text{ mod } 101$$
 ✓

$$2^{16} = (2^8)^2 = 54^2 = 88 \text{ mod } 101$$
 ✓

$$2^{32} = (2^{16})^2 = 68 \text{ mod } 101$$
 ✓ (5 multi)

$$2^{35} = \underbrace{2^{32}}_{2 \text{ more}} \cdot 2^2 \cdot 2 = 68 \cdot 4 \cdot 2 = \underline{39 \text{ mod } 101}$$

Yay! (∴)

2 more

Yay! ☺

Upper: To compute $a^n \bmod p$
requires just $O(\log_2 n)$ many
multiplications mod p .

Big-Oh!

(Positive)
Constant $\times \log_2 n$ many

These kind of tricks are called
"recursive squaring" or "the binary method".

Big-Oh is used for "rough" upper bounds...

To repeat: $a^n \bmod p$ ($n \in \{0, \dots, p\}$)
can be done using $O(\log_2 p)$ multiplications
mod p .

Example: $2^{35} \bmod 101$ took

7 mults 101.

$$\log_2(101) = 6.658\dots$$

$$\Rightarrow \# \text{ mults} \leq 2 \cdot \log_2(101) \\ = O(\log_2(101))$$

In this sense, computing powers mod p is "easy".

However, computing discrete log seems harder!

Given (g, y, p) ,
find x such that
 $g^x = y \bmod p$.

Example: What x gives you $2^x = 52 \bmod 101$?

Example: What x gives you $2^x = 52 \pmod{101}$?
It is $x = 68$. (?)

More generally, to find x , you could try
every $x \in \{1, \dots, p-1\}$ to find $g^x = y \pmod{p}$.

Can you do better? Ⓢ Brute-force: $O(p \cdot (\odot))$
 $= O(p \log^3 p)$

Ⓢ Baby Steps/Giant Steps
(Shanks, 1970s...): $O(\sqrt{p} \log^3 p)$

Ⓢ Index Calculus: $O((\log p)^{1/3} \cdot (\log \log p)^{2/3})$
(Pollard, ..., 1980s, ...)

Ⓢ $O((\log p)^{\text{const}})$? ? ?

(Quantum computers
can go this fast, with a
positive error prob, but you can not
get one yet...)

Diffie-Hellman (1977):

To share a secret, do this:

- ① Pick a large prime p (hundreds of digits, and ...)
- ② For Mr. Rogers and Lady Evelyn to share a
secret, R picks random $a \in \{2, \dots, p-2\}$,
 E ... $b \in \{2, \dots, p-2\}$,

An example

$E \dots$

$b \in \{2, \dots, p-2\}$

③ $R \rightarrow g^a \bmod p \rightarrow E$

$E \rightarrow g^b \bmod p \rightarrow R$

An enemy
only sees p ,
 g^a, g^b !

④ R computes $(g^b)^a = g^{ab} \bmod p$.

$E \dots - (g^a)^b = g^{ab} \bmod p$.

You can find
 g^{ab} if you compute
discrete log...
but that seems
hard!